

Study of OWASP Top 10 Vulnerabilities in Software

AIM

Study of OWASP Top 10 Vulnerabilities in software.

SOFTWARE REQUIRED

- Computer or laptop
- Web browser (Google Chrome, Mozilla Firefox, etc.)
- OWASP Top 10 documentation
- Web application testing environment (optional)

THEORY

OWASP (Open Worldwide Application Security Project) is a non-profit organization that provides resources and guidelines for improving software and web application security. The OWASP Top 10 is a widely used awareness list of the most critical security risks in web applications.

The OWASP Top 10 vulnerabilities are:

1. Broken Access Control – Allows users to access resources or functions without proper authorization.
2. Cryptographic Failures – Occur when sensitive data is not properly protected using encryption.
3. Injection – Happens when untrusted input is interpreted as commands or queries by an application.
4. Insecure Design – Results from security weaknesses in the design or architecture of an application.
5. Security Misconfiguration – Caused by incorrect or insecure configuration of applications, servers, or security controls.
6. Vulnerable and Outdated Components – Using software libraries, frameworks, or components with known security vulnerabilities.
7. Identification and Authentication Failures – Weaknesses in login, authentication, session management, or password handling.
8. Software and Data Integrity Failures – Occur when software updates, code, or data are not properly verified for integrity.
9. Security Logging and Monitoring Failures – Inadequate logging and monitoring can make attacks difficult to detect and investigate.
10. Server-Side Request Forgery (SSRF) – Allows an attacker to make a server send requests to unintended internal or external resources.

Studying these vulnerabilities helps developers identify security risks early and implement

secure coding practices, proper authentication, access controls, encryption, secure configuration, regular updates, and effective monitoring.

CONCLUSION

The study of the OWASP Top 10 helps in understanding the most important security risks affecting web applications. By identifying these vulnerabilities and applying suitable security controls, developers can build safer and more secure software.